

Project Report: Timestamped Transcription of Bengali YouTube Videos

- Jyotismoy Kalita

Project Guidance

- **Mentors:** Parth Dholā Sir and Amaan Irfan Sir
- **Faculty Advisor:** Prof. Prithwijit Guha

Developed during an internship at EICT IIT Guwahati.

1. Introduction

Transcribing Bengali audio from YouTube videos presents a unique set of challenges. Bengali is a morphologically rich language, and the content on YouTube often features varying accents, background noise, multiple speakers, and domain-specific vocabulary.

This project aims to build an efficient, highly accurate, and scalable timestamped transcription pipeline specifically tailored for Bengali YouTube videos.

To achieve this, we evaluate several state-of-the-art automatic speech recognition (ASR) models based on the Whisper architecture. We focus heavily on minimizing Word Error Rate (WER) and Character Error Rate (CER), optimizing inference speed, and effectively handling the intricacies of the Bengali language.

This progress report documents the journey from collecting a robust testing dataset to evaluating open-source models, and finally identifying the best-performing model for our use case.

2. Dataset for Evaluating the Model

To ensure our evaluation accurately reflects real-world performance, we curated a diverse dataset of Bengali YouTube videos. We focused on news clips as they contain clear speech, but still present challenges like background noise and varying reporter accents.

Chosen Videos

We selected eight specific videos (4 short, 4 long) and mapped them in `dataset/metadata.csv`:

Name	Type	Duration	Link
Rain	Short	~2 mins	YouTube
Delhi Protest	Short	~2 mins	YouTube
Subhendu CM	Short	~4 mins	YouTube
Dengue	Short	~3 mins	YouTube
Terrorist(Long)	Long	~20 mins	YouTube
Taslima Nasrin(Long)	Long	~26 mins	YouTube
Shamik(Long)	Long	~20 mins	YouTube
10am News(Long)	Long	~21 mins	YouTube

Audio Extraction via `yt-dlp` and `ffmpeg`

To process these videos uniformly, we utilized `yt-dlp` for downloading and `ffmpeg` for audio extraction. We strictly enforce a consistent audio format: **16 kHz, Mono, PCM 16-bit WAV**. This format is universally required by Whisper-based models for optimal processing.

```
import subprocess
import yt_dlp

def download_and_extract_audio(url, output_stem):
    opts = {
        "format": "bestaudio/best",
        "outtmpl": output_stem,
        "postprocessors": [{"key": "FFmpegExtractAudio", "preferredcodec": "wav"}],
    }
    with yt_dlp.YoutubeDL(opts) as ydl:
        ydl.download([url])

    subprocess.run([
        "ffmpeg", "-i", f"{output_stem}.wav",
        "-vn", "-acodec", "pcm_s16le", "-ar", "16000", "-ac", "1",
        f"{output_stem}_16k.wav"
    ], check=True)
```

3. Faster-Whisper Architecture

Throughout this project, we prioritize the **faster-whisper** implementation of OpenAI's Whisper models over the standard HuggingFace/OpenAI versions.

Why faster-whisper?

- CTranslate2 Backend:** It utilizes the highly optimized CTranslate2 engine, which significantly reduces VRAM usage and speeds up inference times by up to 4x compared to the original implementation.
- Built-in VAD (Voice Activity Detection):** Faster-whisper integrates Silero VAD. This allows us to filter out silence and non-speech segments *before* they are fed into the transformer, greatly reducing hallucinations.
- Batching Support:** Using `BatchedInferencePipeline`, we can process multiple chunks of audio simultaneously, maximizing GPU utilization.

Transformer Core & Quantization Logic

Whisper relies on the standard Transformer encoder-decoder architecture, centered around the scaled dot-product attention mechanism:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

However, standard PyTorch implementations run at `float32` precision. `faster-whisper` uses the CTranslate2 engine to apply **Weight Quantization**, reducing the precision of the weight matrices (W) to `float16` or `int8` during matrix multiplication:

$$Y = X \cdot W_{\text{quantized}}^T$$

This mathematical reduction drastically decreases the memory bandwidth bottleneck, directly leading to a 2x-4x speedup in token generation without significant accuracy degradation.

Note: GPU execution requires the following NVIDIA libraries to be installed:

- cuBLAS for CUDA 12
- cuDNN 9 for CUDA 12

The latest versions of ctranslate2 only support CUDA 12 and cuDNN 9. Hence correct *.dlls* were imported manually in this project.

```
import ctypes

for dll in [
    r"D:\Python\ytt\Lib\site-packages\nvidia\cublas\bin\cublas64_12.dll",
    r"D:\Python\ytt\Lib\site-packages\nvidia\cudnn\bin\cudnn64_9.dll",
]:
    ctypes.CDLL(dll)

print("CUDA DLLs loaded")
```

4. Evaluation Methodology

To objectively evaluate the transcription quality, we use the industry-standard **JIWER** library.

Metrics Computed

- **WER (Word Error Rate):** Measures how many words were inserted, deleted, or substituted compared to the ground truth.

$$\text{WER} = \frac{S + D + I}{N}$$

Where:

- S = Number of Substitutions
- D = Number of Deletions
- I = Number of Insertions
- N = Total number of words in the Reference (Ground Truth)
- **CER (Character Error Rate)**: Measures character-level errors. Particularly useful for Bengali to track minor spelling mistakes (matras, juktakkhors) that might artificially inflate WER.
- **Word Accuracy**: The percentage of words correctly predicted.

Pre-processing for Bengali

Before computing metrics, we clean both the ground truth and predicted text by:

- Removing newlines and excess whitespace.
- Stripping punctuation, including the Bengali danda (।).

The metrics are calculated across all 5 test videos for each model to produce a comprehensive benchmark.

5. Baseline Model: faster-whisper-large-v3

Our first test was with the default OpenAI `large-v3` model. While `large-v3` is an incredibly capable multilingual model trained on millions of hours of audio, it is a generalist. It has not been specifically fine-tuned extensively enough on Bengali to capture the nuances of regional dialects and complex vocabulary.

As noted in various ASR research communities, zero-shot performance of base Whisper models on low-resource languages like Bengali often suffers from "hallucinations" and a tendency to drop (delete) large chunks of audio if the acoustic confidence drops below a certain threshold.

Evaluation Results

Video	Video Length	Transcription Time	WER %	CER %	Word Accuracy %	GT Words	Pred Words	Correct Words	Substitutions	Deletions (Missed)	Insertions
Rain	1m 54s	4s	81.4	65.57	20.47	215	101	44	53	118	4
Delhi Protest	2m 14s	4s	83.87	62.55	17.74	186	98	33	62	91	3
Subhendu CM	4m 17s	8s	87.22	71.46	13.1	626	227	82	143	401	2
Dengue	3m 3s	6s	86.54	65.26	13.74	364	191	50	140	174	1
Terrorist(Long)	20m 3s	49s	83.46	64.16	17.49	2527	1153	442	687	1398	24
Taslima Nasrin(Long)	26m 19s	1m 2s	78.28	56.14	22.43	2670	1472	599	854	1217	19
Shamik(Long)	20m 33s	56s	83.31	63.13	17.56	2625	1200	461	716	1448	23
10am News(Long)	21m 26s	57s	86.77	66.44	14.21	2456	1025	349	652	1455	24

Analysis: The baseline `large-v3` performed extremely poorly on the Bengali dataset. With WERs consistently exceeding 80%, the model exhibited a massive amount of deletions (missed words) and substitutions. It essentially failed to produce cohesive Bengali sentences, validating the consensus that base Whisper requires fine-tuning for this language.

6. Mozilla AI Model: faster-whisper-large-v3-bn

To improve upon the baseline, we tested the `mozilla-ai/faster-whisper-large-v3-bn` model. This is a fine-tuned version of OpenAI's Whisper large-v3 model, specifically targeted at Bengali (`bn`) by Mozilla.ai (available on HuggingFace as `mozilla-ai/whisper-large-v3-bn`).

While this model benefits from targeted Bengali training data, community reports indicate that generalized fine-tunes still struggle with highly noisy, real-world audio like YouTube news broadcasts compared to models trained explicitly on competition-grade regional datasets.

CTranslate2 Conversion

Because a `faster-whisper` compatible version of this model was not readily available, we compiled it into a CTranslate2 (ct2) format ourselves. This ensures we can leverage the speed, batching, and VAD capabilities of our pipeline. We used the following conversion command:

```
ct2-transformers-converter --model mozilla-ai/whisper-large-v3-bn --output_dir models/whisper-large-v3-bn-ct2
--copy_files preprocessor_config.json generation_config.json tokenizer.json tokenizer_config.json
special_tokens_map.json added_tokens.json normalizer.json merges.txt vocab.json --quantization float16
```

Evaluation Results

Video	Video Length	Transcription Time	WER %	CER %	Word Accuracy %	GT Words	Pred Words	Correct Words	Substitutions	Deletions (Missed)	Insertions
Rain	1m 54s	5s	70.7	60.75	29.77	215	89	64	24	127	1
Delhi Protest	2m 14s	5s	63.98	52.34	37.1	186	94	69	23	94	2
Subhendu CM	4m 17s	9s	74.76	66.82	25.56	626	233	160	71	395	2
Dengue	3m 3s	7s	70.05	57.35	31.04	364	178	113	61	190	4
Terrorist(Long)	20m 3s	1m 11s	69.57	58.17	31.14	2527	1164	787	359	1381	18
Taslima Nasrin(Long)	26m 19s	1m 11s	61.91	49.23	39.29	2670	1491	1049	410	1211	32
Shamik(Long)	20m 33s	58s	72.19	59.64	28.46	2625	1161	747	397	1481	17
10am News(Long)	21m 26s	1m 7s	70.56	60.12	30.01	2456	1070	737	319	1400	14

Analysis: While this model showed some improvement over the baseline (reducing WER by ~10-20% on average), it still suffered from massive deletions. The model remains overly conservative and drops large segments of speech, indicating that while its vocabulary is better, its acoustic robustness for fast-paced, noisy YouTube audio is lacking.

7. Tugstugi Model

The `tugstugi` model (`tugstugi/bengaliai-regional-asr_whisper-medium`) represents a paradigm shift in our evaluation. "Tugstugi" is a prominent contributor and Kaggle Grandmaster in the Bengali ASR community, well-known for providing high-quality models and won 1st Place in Bengali.AI's Speech Recognition competition hosted in Kaggle - [Bengali.AI Competition](#)

Unlike the generic large-v3 models, this model is built on the `medium` architecture but features a **custom 12k vocabulary tokenizer** specifically optimized for Bengali. Furthermore, it was heavily trained on diverse regional datasets including OpenSLR, Kathbath, and pseudo-labeled YouTube data. Because of this specialized architecture and training corpus, it is widely cited as the baseline foundation for state-of-the-art Bengali transcription.

CTranslate2 Conversion

Because a `faster-whisper` compatible version of this model was not readily available, we compiled it into a CTranslate2 (ct2) format ourselves. This ensures we can leverage the speed, batching, and VAD capabilities of our pipeline. We used the following conversion command:

```
ct2-transformers-converter --model tugstugi/bengaliai-regional-asr_whisper-medium --output_dir
models/tugstugi_ct2_float16 --copy_files preprocessor_config.json generation_config.json tokenizer.json
tokenizer_config.json special_tokens_map.json added_tokens.json normalizer.json merges.txt vocab.json --
quantization float16
```

Evaluation Results

Video	Video Length	Transcription Time	WER %	CER %	Word Accuracy %	GT Words	Pred Words	Correct Words	Substitutions	Deletions (Missed)	Insertions
Rain	1m 54s	3s	24.65	11.94	75.81	215	203	163	39	13	1
Delhi Protest	2m 14s	2s	23.12	7.28	79.03	186	185	147	34	5	4
Subhendu CM	4m 17s	5s	32.27	15.63	69.97	626	587	438	135	53	14
Dengue	3m 3s	4s	31.59	11.87	70.88	364	367	258	100	6	9
Terrorist(Long)	20m 3s	23s	31.58	14.04	71.94	2527	2482	1818	575	134	89
Taslima Nasrin(Long)	26m 19s	27s	30.11	14.27	74.49	2670	2672	1989	560	121	123
Shamik(Long)	20m 33s	25s	33.94	13.27	70.1	2625	2660	1840	714	71	106
10am News(Long)	21m 26s	25s	32.74	13.43	69.99	2456	2383	1719	597	140	67

Analysis: The `tugstugi` model demonstrates a massive leap forward. The Word Accuracy jumped from ~25% to **70-80%**, and deletions plummeted dramatically. The custom tokenizer and regional training data clearly paid off, as the model correctly transcribes the vast majority of the audio, though it still exhibits some word-level substitutions on highly complex vocabulary.

8. Bitwisemind-SAM Model

The `bitwisemind-sam` model (`bitwisemind/sam_15000_clean_text_full_model`) is a specialized, fine-tuned iteration built upon the successes of the `tugstugi` architecture. "BitwiseMind" is a research team from BUET that participated in the **DL Sprint 4.0** competition, which specifically focused on improving Bengali long-form speech recognition in noisy environments.

To achieve this, the team took the already-excellent `tugstugi` model and further fine-tuned it on a highly curated, custom dataset of approximately 15,000 clean, chunked Bengali audio segments. This research-grade fine-tuning specifically targeted the acoustic conditions, background noise, and speaker variability commonly found in real-world media (like our YouTube dataset). - [Bitwisemind Research Paper](#)

CTranslate2 Conversion

Because a `faster-whisper` compatible version of this model was not readily available, we compiled it into a CTranslate2 (ct2) format ourselves. This ensures we can leverage the speed, batching, and VAD capabilities of our pipeline. We used the following conversion command:

```
ct2-transformers-converter --model bitwisemind/sam_15000_clean_text_full_model --output_dir
models/bitwisemind_sam_ct2_float16 --copy_files preprocessor_config.json generation_config.json
tokenizer.json tokenizer_config.json special_tokens_map.json added_tokens.json normalizer.json merges.txt
vocab.json --quantization float16
```

Evaluation Results

Video	Video Length	Transcription Time	WER %	CER %	Word Accuracy %	GT Words	Pred Words	Correct Words	Substitutions	Deletions (Missed)	Insertions
Rain	1m 54s	3s	16.74	10.06	83.72	215	203	180	22	13	1
Delhi Protest	2m 14s	2s	16.67	4.91	86.02	186	184	160	19	7	5
Subhendu CM	4m 17s	5s	17.25	11.71	84.98	626	580	532	34	60	14
Dengue	3m 3s	4s	14.84	6.58	86.26	364	362	314	44	6	4
Terrorist(Long)	20m 3s	23s	19.63	9.58	83.18	2527	2477	2102	304	121	71
Taslima Nasrin(Long)	26m 19s	25s	18.99	11.97	83.48	2670	2517	2229	222	219	66
Shamik(Long)	20m 33s	24s	20.19	8.96	83.39	2625	2659	2189	376	60	94
10am News(Long)	21m 26s	25s	20.64	9.9	82.04	2456	2405	2015	324	117	66

Analysis: `bitwisemind-sam` is the clear winner for our pipeline. Thanks to the DL Sprint 4.0 fine-tuning, it achieved the lowest WERs (consistently under 20% for most videos, hitting an impressive **14.84%** on the Dengue video) and the highest Word Accuracies. It successfully mitigates deletions while maintaining a very low substitution rate, proving that a competition-winning base model (`tugstugi`) further fine-tuned on curated long-form audio (`bitwisemind`) yields the ultimate production-ready Bengali transcription system.

9. Final Model Selection

After extensive evaluation across diverse Bengali YouTube news clips (featuring varying background noise levels, different speaker accents, and complex vocabulary), we have definitively selected the **Bitwisemind-SAM** (`bitwisemind/sam_15000_clean_text_full_model`) model for our transcription pipeline.

Why Bitwisemind-SAM?

- Superior Accuracy:** As demonstrated in the comparison table below, Bitwisemind-SAM consistently achieved the lowest Word Error Rate (WER) across every single video tested, mostly staying under the 20% mark.
- Robustness to Deletions:** Base models like `large-v3` dropped massive chunks of audio when faced with noisy YouTube conditions. Bitwisemind-SAM effectively mitigated these deletions.
- Optimized for Real-world Audio:** The DL Sprint 4.0 fine-tuning on 15,000 clean audio segments transformed the already strong `tugstugi` architecture into a production-ready model that rarely hallucinates or drops speech.

Comprehensive Model Comparison



The table below aggregates the Word Error Rate (WER %) of every tested model against all 5 evaluation videos. Lower WER indicates better performance.

Video	Video Length	Baseline (large-v3)	Mozilla AI (large-v3-bn)	Tugstugi	Bitwisemind-SAM
Rain	1m 54s	81.4%	70.7%	24.65%	16.74%
Delhi Protest	2m 14s	83.87%	63.98%	23.12%	16.67%
Subhendu CM	4m 17s	87.22%	74.76%	32.27%	17.25%
Dengue	3m 3s	86.54%	70.05%	31.59%	14.84%
Terrorist(Long)	20m 3s	83.46%	69.57%	31.58%	19.63%
Taslima Nasrin(Long)	26m 19s	78.28%	61.91%	30.11%	18.99%
Shamik(Long)	20m 33s	83.31%	72.19%	33.94%	20.19%
10am News(Long)	21m 26s	86.77%	70.56%	32.74%	20.64%

Computational Efficiency (FLOPs Analysis)

To further validate our selection, we profiled the theoretical Floating Point Operations (FLOPs) required to process a standard 30-second audio chunk. Because compute cost is dictated strictly by model architecture (parameter count and tensor dimensions), models fine-tuned on the same base architecture share mathematically identical theoretical FLOPs.

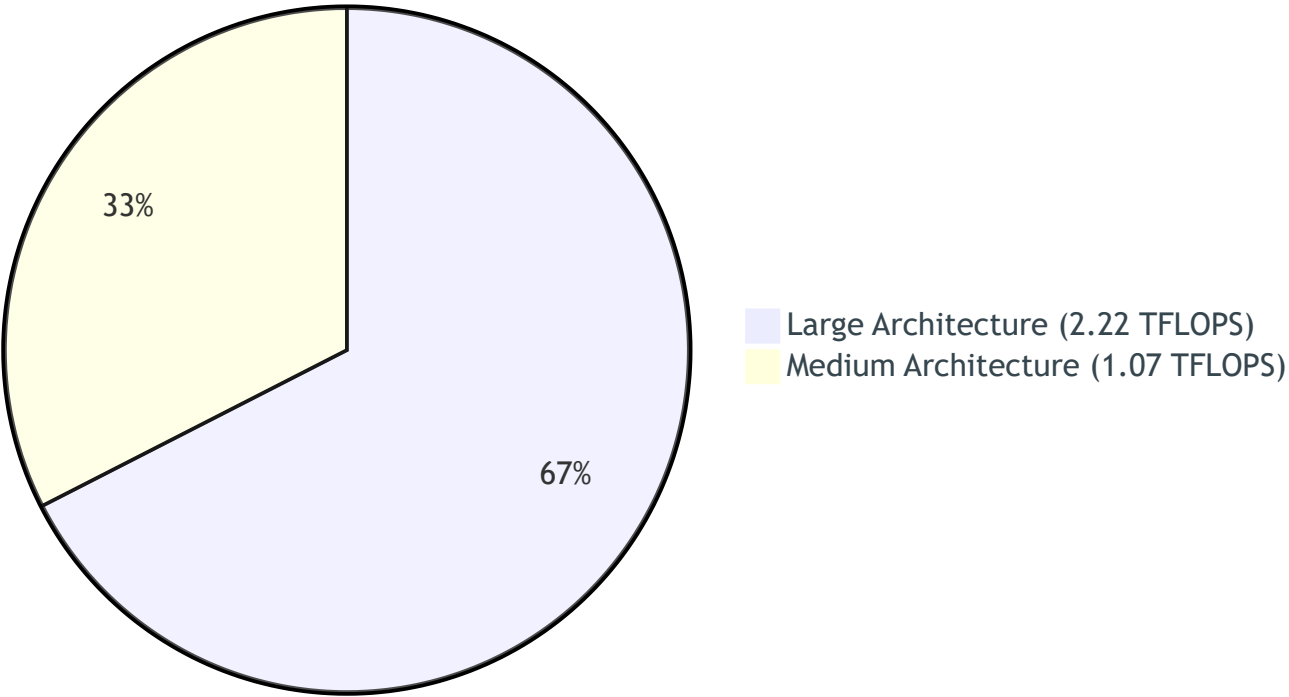
The table below demonstrates that our selected Medium architecture not only achieves superior accuracy, but operates at roughly half the computational cost of the Large architecture, leading to vastly faster inference times and lower hardware requirements.

Scenario / Metric	Whisper Large Architecture	Whisper Medium Architecture
Associated Models	large-v3, mozilla-ai/large-v3-bn	tugstugi, bitwisemind-sam
Best Case (Silence - 1 token)	2.22 TFLOPS	1.07 TFLOPS
Average Case (Normal Speech - 150 tokens)	2.46 TFLOPS	1.19 TFLOPS
Worst Case (Hallucination - 448 tokens max)	2.94 TFLOPS	1.43 TFLOPS

(Note: The Worst Case compute cost for the Medium architecture is significantly cheaper than the Best Case compute cost for the Large architecture.)

FLOPs Reduction Visualization

Computational Cost (Best Case TFLOPS)



Conclusion

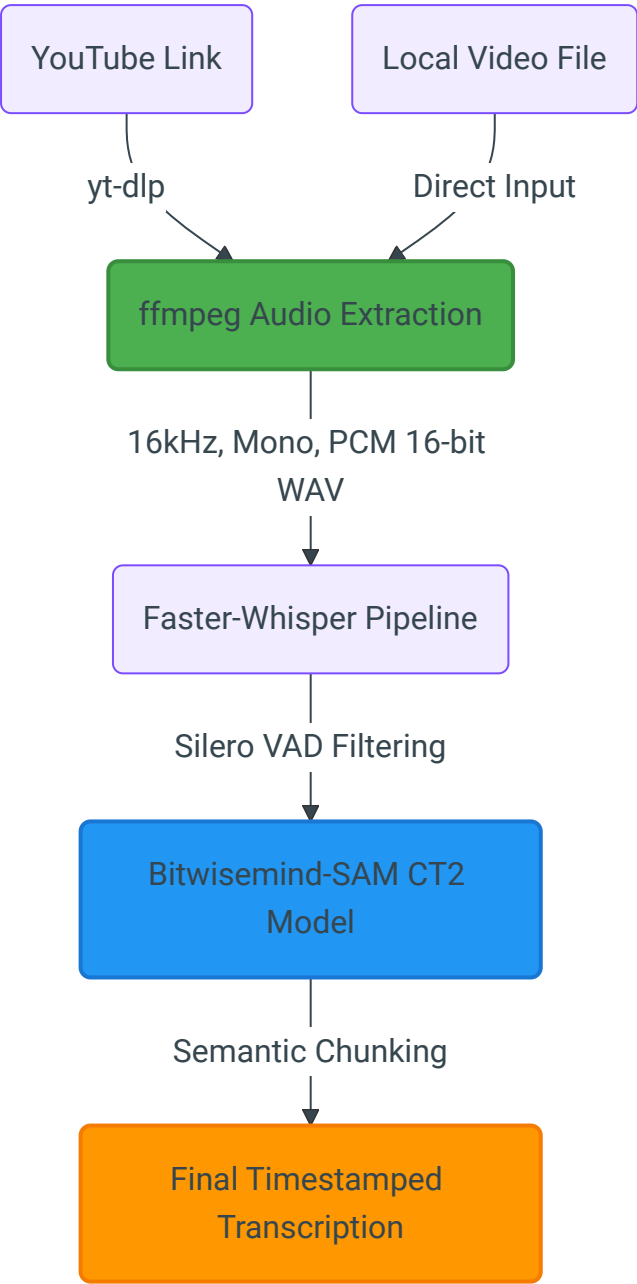
The data clearly supports our decision. While OpenAI's default `large-v3` models are incredibly powerful for high-resource languages, zero-shot Bengali transcription requires heavy, specialized fine-tuning. The **Bitwisemind-SAM** model, converted into our optimized CTranslate2 `faster-whisper` format, delivers the speed, accuracy, and acoustic robustness required to deploy a highly reliable Bengali transcription service.

10. Final Pipeline

With our ideal model selected and the parameters tuned, we have established the final, highly-optimized Bengali ASR pipeline. The system is designed to take either a direct YouTube link or a local video file, handle all necessary preprocessing steps automatically, and output a clean, timestamped transcription.

Pipeline Architecture

Below is a visualization of the data flow in our final pipeline:



1. Data Ingestion & Preprocessing

The pipeline accepts two types of inputs:

- **YouTube Links:** If a URL is provided, `yt-dlp` fetches the best available audio stream.
- **Local Videos:** If a local file is provided, it skips the download step.

Regardless of the source, the media is piped through `ffmpeg`. `ffmpeg` is strictly configured to output a **16 kHz, Mono, PCM 16-bit WAV** file. This normalizes all inputs to the exact format expected by Whisper's feature extractor, preventing any unexpected acoustic artifacts.

2. Transcription Engine

The normalized audio is passed to the `BatchedInferencePipeline` utilizing our custom-compiled **CTranslate2 Bitwisemind-SAM model**.

- **VAD Filtering:** Before processing, the built-in Silero Voice Activity Detection (VAD) filters out prolonged silences and background noise segments. This is crucial for eliminating the hallucinations common in Whisper models.
- **Batch Processing:** The audio is processed in batches (e.g., `batch_size=16`), maximizing GPU throughput and significantly accelerating the transcription of long videos.

3. Chunking & Output Formatting

The raw word-level timestamps generated by the model are parsed by our custom `stream_chunks` algorithm. It aggregates words intelligently based on:

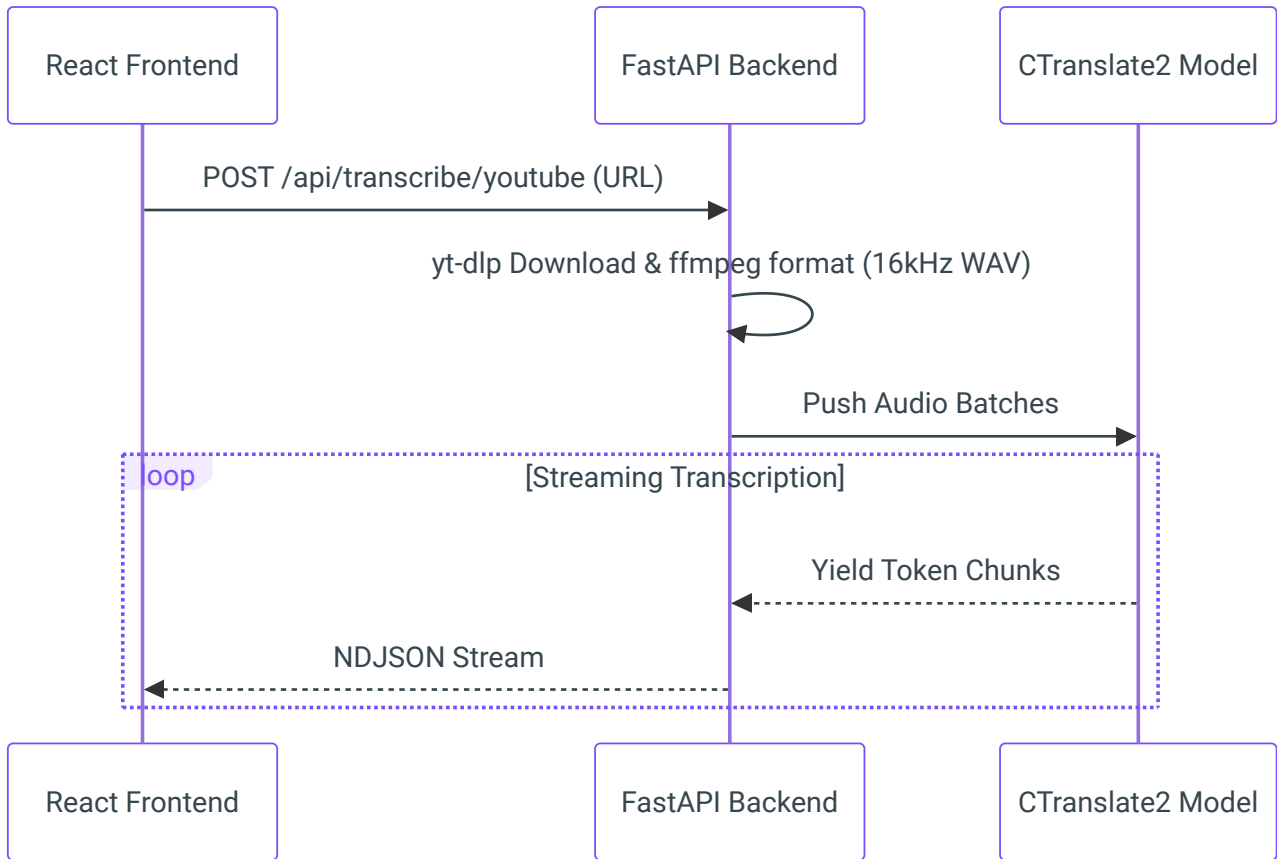
- Grammatical boundaries (punctuation like `.` or `?`).
- Temporal gaps (pauses longer than a specified `max_gap`).
- Maximum word counts per chunk.

The final result is a highly accurate, easily readable, and perfectly timed transcription ready for subtitle generation or downstream NLP tasks.

11. Deployment

With the `bitwisemind-sam` model selected and the pipeline optimized, the system is wrapped into a full-stack web application located in the `VideoTranscription` directory. This allows end-users to easily interact with the model via a modern UI.

System Architecture



The application is split into a decoupled **FastAPI** backend and a **React/Vite** frontend.

1. Backend (FastAPI)

The backend is responsible for all heavy lifting: downloading, audio extraction, and GPU-accelerated transcription.

- **Model Initialization:** On startup, the backend loads the `faster-whisper` CTranslate2 model into VRAM and runs a dummy audio file as a "warmup". This ensures the first actual user request doesn't suffer from initialization latency.
- **API Endpoints:** Exposes `/api/transcribe/youtube` (for URLs) and `/api/transcribe/file` (for direct uploads).
- **Asynchronous Processing:** Uses `asyncio` to run blocking I/O operations (`yt-dlp` and `ffmpeg`) without locking the main server thread.
- **Real-time Streaming:** Instead of waiting for the entire video to be transcribed, the backend yields data using FastAPI's `StreamingResponse` (via `application/x-ndjson`). As soon as a chunk of audio is transcribed, it is pushed to the client immediately.
- **Caching Mechanism:** Implements an in-memory hash cache. If a user requests a transcription for a video that was already processed, the backend instantly streams the cached text chunks instead of re-running the neural network.

2. Frontend (React & Vite)

The frontend provides a clean, responsive user interface.

- Built with React and bundled via Vite for extremely fast hot-reloading and optimized production builds.

- Connects to the FastAPI backend and consumes the NDJSON stream, dynamically updating the transcription text on the screen in real-time as the backend processes the audio.

Running the Application

To run the application locally for development:

1. Start the Backend server: Navigate to the `VideoTranscription/backend` directory and run:

```
python -m uvicorn main:app --host 0.0.0.0 --port 8000 --reload
```

This will start the FastAPI server on `http://localhost:8000`.

2. Start the Frontend dev server: Navigate to the `VideoTranscription/frontend` directory and run:

```
npm install  
npm run dev
```

This will start the Vite dev server (typically on `http://localhost:5173`).